

Karl Friston talk, 2021 IWEI HD 720p

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so with that um carl if you're here hopefully you're here uh please feel free to uh unmute yourself and uh and share your screen welcome thank you very much thank you for another introduction let me share my screen and then great well it's a great pleasure to be here and also nice to see so many so many old friends as well that's a perverse gift of covid and i was torn um about what to talk about over the next half hour between looking at markov blankets as one way of studying emergentism and embodiment when the internal states synchronize with the external states or focusing on the intelligence part of embodied intelligence but i've elected to go for the intelligence part and pick up on um one of the themes that the first keynote um uh introduced this notion of intrinsic motivation and a theme which came back a number of times particularly in in gentani's and and andy clark's uh presentations the you know the notion of planning and temporal depth and an understanding of the imperatives for um behavior and active engagement with the world that looks into the future so i'm going to focus on the intelligence bit of embodied intelligence and i'm going to cast that in terms of self-evidencing briefly rehearse what i mean by that in terms of normative models of action and perception and possibly uh consciousness and cognition um how it relates to or how self-evidencing um brings to the table some things that would be useful for understanding embodiment in artificial intelligence and then if we have time um just use a um a toy simulation just to unpack how the principles work and some of the fundamentals of the behavior that ensues so i'm taking my lead from the notion that all we need to know in terms of explaining and self-organized behavior of a sentient and possibly intelligent sword um is that everything uh in our brains or in our body and in particular those bits of the body that change the way that the brain and the body samples its its sensorium and its echo niche everything is just in the service of maximizing the evidence for a artifact or a creatures generative model of its world m here so i'm just articulating that mathematically everything is in the service of maximizing probability of some observable outcomes some sensed outcomes given a model of those outcomes and i'm just um portraying here a number of different takes on that that would depend upon your commitments

and your training you could um interpret the log of the probability of those outcomes that you're trying to solicit or behave in a way to maximize the probability of those outcomes being elicited as value the valuable outcomes and you could look at that from the point of view of reinforcement learning optimal control theory or indeed if you're an economist expected utility theory i've cast value um in terms of a negative potential um that is in fact the self information in information theory so uh sometimes referred to as surprise or surprise and that's nice because if we're trying to maximize value or have a normative take on value maximization then what we're also saying is that we also have an information theoretic understanding of the imperatives that um are convinced by self-evidencing in terms of uh principles of minimum redundancy or maximum efficiency and indeed the free energy principles so um just noting that the free energy that some people have mentioned provides a bound on surprise or value when cast in this way in turn that is also nice because the time average of surprisal or self-information is entropy so what we're saying is in self-evidencing then we are just trying to resist the dispersion uh according to the second law uh which of course is the holy grail of self-organization particularly in things like cybernetics and synergetics of course if you're a physiologist it's just homeostasis keeping physiological outcomes or observable states of the body within viable physiological bounds i'm going to actually take another perspective though on this quantity which is the statistician's perspective name of the probability of some outcomes or observables given a model constituting the marginal likelihood of that model or the model evidence and then one can spin off the base in brain hypothesis evidence accumulation predictive coding and the like so that's where i'm coming from um i want to now just drill down on what that means mathematically and it's related to a couple of things that we've already been talking about so why self-evidencing well this this um quantity um constitutes the sort of you know the core part of this free energy in machine learning this is known as an evidence lower bound why well because there's a bound here which can never be less than zero which means that if i maximize f i am necessarily maximizing a bound on the log evidence hence self-evidencing um this quantity is going to be interesting later on but let me just focus in the first instance of why it will be useful from the point of view of say artificial intelligence um to maximize the evidence for a model of the world and implicit in having a model is an explainability so the quantity the objective function that i'm trying to optimize is an explicit functional of a model that renders it inherently explainable so we have a transparency and expandability and indeed interpretability for free that leads on to something quite important which is optimal bayesian design um and of course

the interpretation of self-evidencing in terms of a bayesian brain or a prediction machine a constructive organ making inferences about its environment um and of course when you embody or put that in in an active context we have um a degree of self-assembly autophasis and self-teaching or autodidactic um capacitors which leads to a principled way of understanding data foraging and epistemics and possibly animal or generalized artificial intelligence but i want to just rearrange that equation which is the way that somebody machine learning might understand um this free energy bound um i'm just moving the terms around just to express it in a way that a statistician would be more familiar with it which is basically the imperative to maximize the accuracy for your explanations or predictions of sensed outcomes whilst at the same time minimizing the complexity where the complexity is the degree of belief updating or the amount you have to change your mind in relation to new data the degrees of freedom you're using up in terms of providing an accurate explanation and i'm going on about that because i was intrigued by the discussion of complexity either in the brain or in the body and simplicity and the like and this is a central notion in terms of self-evidencing where if not um most of but certainly a substantial part of the drives to maximize this evidence lower bound is in getting the simplest most parsimonious explanations implicit in your generative model um that you can so that's just across the statement of occam's principle and you see that emerging in structure function relationships in terms of things like modular factorizations in computers or indeed the brain where it's known as functional specialization efficient coding it also um [Music] provides an interesting um take on bounded rationality that you know it's not a question of just being rational and very accurate it's a way it's a special kind and optimal maximum evidence way of being rational where you're trying to find the simplest explanations so it easily accommodates the notion of heuristics via prize inherent in the genitive model namely the probability of some states of affairs in the world generating uh these outcomes here um it also practically means that when the artifact is behaving in an optimal way in relation to free energy or marginal likelihood there's some simple physics which says it should be doing it in the most thermodynamically efficient way so you can invoke the jninski equality and landauer's principle to say that if you've got the right solution or your artifact is self-evidencing optimally then it will be consuming the least amount of electricity it will be as small and as simple as possible and so that's the basic if you like motivations for self-evidencing cast in terms of maximizing model evidence or marginal likelihood what i want to do though is ask what would that look like if i had to choose how to use my body to maximize the free energy in the future and essentially what i'm going to be

saying is that if i condition the outcomes yet to be observed on some plan then accuracy and complexity translate into two things which yeah which we will all immediately recognize basically um they're going to translate into an optimal design and an optimum decision making so what the story i'm telling here is that self evidencing just is a mixture of base optimal design in the way that we go and solicit evidence from the world evidence for our own existence for our generative model under pry beliefs about the kind of thing that i am so here in this instance um the base optimality of is of a decision theoretics or that if i choose to do this then the outcomes are going to be close to my prior beliefs about the kind of a thing like me will expect to encounter so i'm just going to briefly rehearse that story um by comparing and contrasting um conventional rl like approaches to optimality versus this more general self-evidencing approach and and i normally start this this lecture um by asking you to imagine that you're a bird of prey and you're hungry and then i asked the audience normally um what are you going to do and they normally say quite correctly i'm going to search and fly around until i find my prey and then i'm going to eat my prey and in that answer there is buried um an important or several important takes on the kind of functions or functionals that you would use to try and describe good behavior so we could either assume the existence of a value function of some states of the world that are contingent upon some action or control variable you here and if this value function of states of the world existed then i could uh optimize the value by selecting the best state action policy π policy by selecting the action u for every given state that maximize this value function and this is known as the optimality principle um however just in saying i'm gonna search for my food before i eat it you're saying something quite profound because searching is a action that resolves uncertainty and uncertainty is an attribute of a probabilistic or a bayesian belief and therefore the thing that you are optimizing can't be a function of unknown states for the world it has to be a function of beliefs about states of the world so in in the sort of bayesian um interpretation i'm denoting posterior or approximate posterior beliefs about states of the world by q so what we're saying is optimal action depends not upon states of the world but on beliefs about states of the world and we get this sort of functional form here the other thing um that the sort of um the owl example um shows is that the order matters it matters whether i first of all search for my prey and then eat it or try to eat it and then search for it which means that there does not exist an optimal state action um policy but more an optimal sequence of behaviors given what i currently believe about the state of the world at this point in time and we can articulate that in terms of a path or a time integral of this um quantity which is in fact going to be um an expected free energy that andy

and a number of other people have already mentioned um leading us to an optimization of a time integral of an energy function which in physics is an action and i'm phrasing it like that because what we've basically got here is a choice between the bellman optimality principle that would seem things like optimal control theory or dynamic programming deep reinforcement learning basic decision theory on one hand or on the other hand we have hamilton's principle of released action um or station reactions more strictly that is exemplified by things like active inference and subsumes things like artificial curiosity and intrinsic motivation that i'm going to cast in terms of optimal bayesian design that optimally resolves uncertainty about my hypotheses about how my body works and how the world works it also entails sequential policy optimization and the distinction between observations of a certain world and partially observed worlds so here's the basic architecture that one has in mind when invoking and we've seen not some lovely examples of this for example giantani's um um illustration this morning um so you know the basic theme um the embodiment gets into the act here in terms of this sort of reciprocal engagement this circular causal link between the inside and the outside that andy was talking about mathematically what what one way in which one can describe this is that you get these outcomes supplied by the environment you use these outcomes to maximize a functional of those outcomes and your beliefs under a genetic model about states causing those outcomes and to form world out there you then use these to roll out effectively into the future under a number of potential policies so this is exactly the poise over possible actions or an action space the action space here is explicitly encoded in terms of sequences of actions or moves on the world denoted by π and each π you can roll out and estimate the expected free energy which i've here decompose into risk and ambiguity and we'll drill down in that onto in that in the next slide at the moment we just need to know now we've got a way of scoring each way forward each one of these possible action um possible policies possible plans can be scored in relation to our beliefs about the kinds of things that we expect to encounter and the kinds of things that we do with our bodies and that score can then be used to find the most likely policy and with selected action from belief about the most likely policy it changes states in the world the world supplies a new outcome and uh off we go again in this perception action-like cycle that has this planning and policy selection um as an integral part of it um so this is the the bit that sort of um links the notion of simplicity and complexity and to the um more prospective aspects of action and planning so what i've done here is rewrite down the the evidence lower bound the free energy here in terms of accuracy and complexity and remember the complexities is the difference between your posteriors and your priors the amount you've

had to change your mind which is the energy hungry part of it and the accuracy here and just by moving the terms around uh we can reconstitute or recover uh the ex the expression that lends its um elbow acronym the evidence lower bound because this quantity um is always greater than zero and here's the expected value has a very very similar form the only difference is that we're now conditioning upon some possible action or possible sequence of actions and interestingly what happens now is of course you don't have the observations that would be used to evaluate the free energy because they are now random variables in the future so you now have to take your expectation over the prediction of the outcomes in the future given i'm going to do that and hence expected free energy but the form is the same and i like this because it's really quite revealing so the complexity becomes risk the accuracy becomes on the negative accuracy becomes which means that we're going to choose those things if we are choosing those things that maximize expected free energy in the future we are necessarily going to be minimizing our risk and minimizing our ambiguity and i'll explain what that means in a second or we can carve up this expected free energy in another way in terms of the intrinsic and the extrinsic value and this is the intrinsic motivation so we have basically um both the complexity and the intrinsic motivation um or the expected complexity and intrinsic as different ways of carving up this underlying imperative for self-evidencing but where you're maximizing the expected evidence in the future given a particular action just to unpack this a little bit for those people in different fields if i ignore the extrinsic value at the moment and by which i mean i've got some prior beliefs about the kind of outcomes that i as a creature will expect to encounter if i have no particular preferences prior preferences then we're just left with this thing here which is the expected evidence bound and this quantity here is just the difference in my posterior beliefs with and without observations that i would get if i pursued this policy and of course this is um the basis of the mutual information between the consequences and their causes states here and that will be expected under that particular policy if you do visual search or active sensing active vision um then you may recognize this as the expected base in surprise basically the amount of information the epistemic affordance attributed or afforded by this particular policy the reduction in certainty about states of affairs in the world out there in the future um given observations um and evidence for those states um uh in the future if i committed to that particular policy um if i take uncertainty of the various kinds off the table we're actually going to get back to reinforcement learning or expected utility but let me just do that carefully let's first of all assume that we've got a simple world in which we can sense the states of the world so we remove the uncertainty about um that's

inherent in the partially aspects of certain models so that we magically our outcomes now become the hidden states of of the world we can see everything in this instance the s's become the o's um and we're just left then the ambiguity goes where there is no more ambiguity in the world we know exactly what's going on um and that just leaves the risk so what is a risk well it's the against the kl divergence between what i expect to happen under this policy and my prior preferences so when i minimize risk what i'm doing is i'm choosing those policies that will lead to states of the world or their outcomes that a priori i prefer so i'm trying to minimize the degree of divergence or difference between again posteriors about future outcomes and price over future outcomes so very very closely related to complexity the complexity cost but now we're talking about the future if i remove the final kind of uncertainty from the table um what we're left with is effectively um and the final kind of uncertainty is that the um the uncertainty due to different relative uncertainty just disappears i'm just left with this thing here which um i set up in the first slide as being really a version of um um expected utility theory where um the the um expected or extrinsic value is just the expected value or the log probability of preferred outcomes under what i anticipate uh to ensure given a particular um given all um policies or actions here so what we've done here is is derive from the simple notion of self-evidencing the service of garnering evidence for your own existence as implied by your model your embodied model of the world and carving it up into two different aspects of bayesian good behavior first of all um i'm going to um commit to those behaviors that would um um or choices that have the greatest expected value and in in statistics that's known as making bayes optimal decisions under some loss function but here we're casting the loss function of the utility function uh in terms of a prior preference and then which is adding to that effectively the information gain um that ensues from pursuing a particular policy and together they constitute the expected free energy or the expected evidence uh afforded by a particular policy so i'm going to close now just by very quickly taking you through some uh embarrassingly simple simulations and we looked at the beautiful simulations uh this morning uh um we we are nowhere near and uh able to uh scale up to that level of sophistication however i think they're still illuminating um um you know in the sense that there's a certain simplicity in you know in what i'm saying even though some of the maths may look a bit complicated and i'm not going to go through this in any detail i want to show you the the the kind of behavior the sort of um that is mandated under this um optimization of expected free energy we normally use markov decision processes sort of discrete state space models of a world graphically cast in terms of transitions from one state to the next point in time

generating outcomes the likelihood mapping where the transitions are dependent upon a policy uh be use your code encoding the um the probability transitions and then we have interesting um well the policy clearly depends on the prior preferences included by this preference or cost function here um we also have in many of the simulations uh um the precision of our beliefs about policies and i just mentioned that because that's the w in john's talk that sort of tunes the hierarchical depth or the ability to learn versus to teach uh depending upon the confidence that a priori you place in your uh your um inactive policies um when interrogating or queering uh querying the row the world a couple of hyper parameters about initial states and when we plug this very generic um form for the generative model into standard elbow or free energy optimization schemes we get a a series of belief updating uh equations that look very much like what the brain does and indeed we know that because of all the um all the simulations and interpretations process theories that we've already um uh witnessed um this morning and in the literature um so you know there aren't many unknowns here we just don't we don't know about the state of the world we don't know what we're doing and we don't know how confident we are in what we're doing um we actually also don't know um the parameters of the of the of the model itself um um you know the the likelihood mappings and the uh the transition matrices so those can be learned um so we just have perception which is resolving uncertainty or inferring states of the world policies select policy selection so having the right poise if that's correct abuse of and its philosophical term on the space or possible actions and and also having a um a kind of second order belief about the the our beliefs about those actions in terms of the confidence or the precision that incidentally looks as if it's encoded by dopamine in human brains um then we have learning of the unknown structure of the world um that would be a particularly interesting from the point of view of the sort of you know the embodiment part of this equation and then action selection is selecting an action from my beliefs about what i do and when i simulate that and i'm going to simulate that um by applying those equations to a little mouse who um can um has to solicit a baited reward from the right or the left um arms uh it's only um it only has two moves and has to stay in one or the other arm once committing to one of those arms but it has a choice of using up one of its moves to go and get an instructional cue that will tell it um definitively whether the reward is there or there so this little rat has the choice of taking a gamble uh and it could be 50 percent correct and staying there um or spending one of its um exploitative moves in an exploratory sense to go resolve uncertainty about the world and then go and secure the um the cheese and and of course this is set up this sort of two-step paradigm is set up just to demonstrate even though the expected

utility of both policies um is identical uh on average um the the self-evidencing uh creature will clearly go and resolve its uncertainty responding to the epistemic affordance of the intrinsic motivation go and get the cue and then uh go and enjoy and exploit it's it's um it's knowledge um um by going to the the arm that is baited and indeed that's what the agent does but what this simulation um is showing is is a systematic and principled and completely non-determined uh non-um stochastic this is a deterministic optimization of expected free energy which means that if you actually introduce a predictability and a structure and a consistency uh to the world which we've done here just by leaving the reward on the left arm perpetually after the first couple of moves then the creature the the rat will come to learn this and therefore salience or the novelty of the uncertainty resolving q slowly wanes and there is no uncertainty after a while because the rat knows that the reward is on the side and therefore the epistemic affordance the um the explorative drive's intrinsic motivation for going to the condition stimulus or the instructional cue starts to disappear and at some point when the intrinsic and the extrinsic well well the intrinsic value of falls below the pragmatic or the extrinsic value it'll just switch to a different kind of behavior and this is incredibly systematic and you see this in many many simulations and indeed in real world uh situations where you know your first imperative is to resolve uncertainty about this particular world this eco-niche this body um and when you've done that you start to then turn to your prior preferences um because there's no more uncertainty to hoover up um so by move 20 the rat just goes straight there quite happily and will do so forever if nothing else changes this is the final slide and um i've got 30 seconds left um so um i put this here just to remind myself um about that initial um ambivalence of about whether to talk about sort of uh the emergence of life-like uh behavior um from first principles or whether to talk about the sort of the mechanics of planning and uh deep charging models of my own action um it would be nice to to to actually connect the two perspectives um and i'm just putting this up as an outstanding challenge uh and the framing which i think that challenge can be met um i won't go into this but other than to if anybody's interested provide a sort of starting point for discussion so here's what we've just been basically saying we can understand the right kinds of in terms of selecting the right possible actions and what are the right ones whether they're the ones that underwrite our self-evidencing that minimize risk and and they have this functional form there's another way of understanding the universe in terms of um non-equilibrium steady states that are normally described in terms of integral fluctuation theorems if you did chemistry or or physics and they have um a very similar structure in terms of articulating the probabilistic development of um states and and and

outcomes consequences upon different states um in terms of chaos divergences to provide or inequalities underwrite much of statistical thermodynamics interestingly you can use the integral fluctuation theorems to derive this if you make one particular assumption or commitment one small move here and all i'll say about this is it hasn't actually been done yet but it does involve some probabilistic relationship about between the probability of a path of action into the future and the ambiguity or the um the negative um the negative entropy of outcomes given their causes um so again linking action and uncertainty you know in an intimate way which may be the key for uh understanding the kinds of systems eye creatures like us that distinguish ourselves from things like viruses or other self-organizing systems that don't plan anyway um i'll leave the last word with einstein everything should be made as but not simpler and all that remains is for me to thank uh those people whose ideas i've been talking about of course thank you for your attention thank you very much indeed i'll stop my sharing thank you very much carl i i'm sure the 348 participants if they could they would unmute and join me in thanking you there's a couple questions arriving in the discord um so before we we get to uh one or two questions um i just like to ask the organizers if they could help connect professor fristen with the discord if he's not there already there's a question from dana damian beautiful theory thank you um does this theory also explain the emergence of belief in the first place um in an elemental way yes um so that would be part of the story that i couldn't tell so um if you just take any um system that has self-organized some kind of non-equilibrium steady state so it persists there are recognizable characteristics and then and then you partition that system into inside states that constitute the inside of a system and the environment and then they're enshrouded by blanket states this is a markov blanket partition then you can use the necessary properties of non-equilibrium steady states solving the density dynamics to show that there is an information geometry where the physical inside states come to parameterize bayesian or probabilistic beliefs about the outside states and that licenses then an information geometry which is a geometry of belief state so if you if you permit me to interpret beliefs as bayesian or probabilistic beliefs conditional probability distributions then mathematically you're saying is there an information geometry there and what i would say yes there is under the condition you've organized to a non-equilibrium steady state great thank you again more questions arriving there's a question here in zoom chat from from andy um if minimizing expected free energy delivers simple planning what delivers deliberate deliberate planning that seems different yet related um i think the distinction is is a very subtle one um and um we don't have time to to get into it because it is a really interesting question so notice

that um i was talking um i was just carving this um functional this sort of expected free energy model evidence into two bits that literally decompose because we're in long space into sort of an information gain effectively and a you know a pragmatic or um expected utility term that added that that would be could be resolved independently so you could argue well look yeah that's fine but that's not really resolving uncertainty about what matters to me in relation to my prior preferences yeah so yeah i'm just resolving all uncertainty irrespective of you know what what i actually think i should be uh the kinds of things i'm going to i'm going to solicit so if andy's asking that question to ask well when does it get more interesting one could algorithmically say well you have to move to sophisticated inference where you actually have to um instead of just resolving uncertainty you have to actually encode what would i believe if i knew that and what would those new beliefs mean as i work towards my preferences it's not difficult to do it's basically a probabilistic generalization of the uh the bellman recursion however i'd actually say you probably don't need to do that because remember although these two bits of the the good ways to choose your policies are literally separable they're linearly separable in log space which means you're actually multiplying so what you are saying in probably in terms of probabilities i am choosing those um policies that have the greatest epistemic affordance given that they afford um outcomes that i prefer so i think there's actually built in for free a certain sort of deliberation there in the sense that you're conditioned upon resolving uncertainty there has to be an aspiration to the goals that are written the prior preferences great thank you um we have some time for discussion at the end of the plenary session so i'm going uh use my host privileges here and ask you one last question um from discord um how does free energy deal with uh groups of agents and a common belief among the group will it promote agreeing uh with the group over further exploration of the environment yeah yes it does it promotes cooperation and generalized synchrony so mathematically you know that that generalized synchrony basically affords a predictability of two agents in a diet or in fact a multi-agent context so if the whole game is to minimize surprise then the best solution is to basically make you exactly like me and make me exactly like you and then our discourse will be in perfect secret we're both mutually predictable and our joint free energy will fall so it's all about finding harmony in groups and a shared narrative a common ground um and from that you can spin up all sorts of hermeneutic interpretations and indeed you know the the imperatives for for language in terms of cultural niche construction under this sort of variational variational frameworks and and that that would have been our third lecture which which you can't do everything in half an hour certainly can't

thank you so much professor fristen on behalf of all the participants here for uh for a fascinating and thought-provoking presentation